

Standardized Knee Meniscus MRI Reporting: An Interdisciplinary Delphi Consensus

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Thirty-three panelists from 23 institutions participated in a three-round Delphi consensus process to establish interdisciplinary guidelines for reporting knee MRI findings on 11 meniscus topics: MRI criteria for tears, possible tears, tear patterns, tear descriptors, intrameniscal descriptors, meniscal roots, posteromedial meniscocapsular junction, meniscus extrusion, assessment of the postoperative meniscus, retear descriptors, and ancillary findings. All panelists completed all Delphi rounds. In round 1, 350 free-text entries (mean, 31.8 entries per topic $\pm$ 2.2 [SD]) were submitted. In round 2, 32 statements (mean, 2.9 entries per topic $\pm$ 1.6) were collected from 11 teams (one orthopedic surgeon and two musculoskeletal radiologists per team), with 75% (24 of 32 panelists) achieving consensus using a 9-point Likert scale. Consensus was defined as a score of at least 7 ($\geq 80\%$ of panelists) and a median score of at least 7 (IQR ≤ 3). In round 3, all statements achieved consensus. Between rounds 2 and 3, agreement significantly increased ($P < .05$) among all panelists on the topics of possible tears (to express uncertainty) and tear descriptors (for treatment planning), among orthopedic surgeons on MRI criteria for tear and the assessment of the postoperative meniscus (specifically on reporting equivocal findings and the use of MR arthrography), and among radiologists on tear patterns, intrameniscal descriptors, meniscal roots, and retear descriptors.

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Knee MRI examinations are frequently performed to supplement clinical examinations and assist in treatment decision-making. However, some terms used in knee MRI reports remain vague or are employed inconsistently. This inconsistency contributes to ineffective communication and a risk of misunderstanding (1). To date, there is a lack of published analyses comparing the interdisciplinary and multi-institutional characterization and reporting of various meniscus findings on knee MRI. Thus, the objective of this study was to identify interdisciplinary consensus guidelines for reporting meniscus findings on knee MRI using a systematic, interactive Delphi approach.

Materials and Methods

The Delphi method is a validated process for achieving interdisciplinary, multi-institutional consensus. This method employs a representative panel of experts, adhering to core principles of anonymity for individual responses and sequential iterations, which provide the opportunity for revised responses (2).

Panelists

A steering committee comprised of four board-certified, fellowship-trained physicians, including two musculoskeletal radiologists (J.C.N., R.D.B.) and two orthopedic surgeons (K.G.S., K.P.S.), reviewed the published literature and identified 11 topics related to the meniscus. These topics included MRI criteria for tears, possible tears, tear patterns, tear descriptors, intrameniscal descriptors, meniscal roots, meniscocapsular junction, meniscus

extrusion, assessment of the postoperative meniscus, retear descriptors, and ancillary findings. Based on this literature review and the specific topics related to the meniscus, the steering committee identified subspecialty experts who are actively involved in national radiologic and orthopedic societies, have conducted scholarly work on the meniscus, and maintain professional practices that handle patients with knee derangements. To maximize the generalizability of the consensus recommendations, the steering committee carefully considered panel diversity, including variations in years of practice, types of clinical practice, patient age groups, and geographic locations.

All 29 subspecialty experts from 23 North American institutions identified by the steering committee accepted the invitation to participate in this Delphi-based consensus-building process. Background information was collected from each panelist, which included four members of the steering committee and 29 subspecialty experts. The panelists were subdivided into 11 teams, with each team consisting of one orthopedic surgeon and two musculoskeletal radiologists. Each team was assigned one meniscus topic and tasked with developing one or more statements accompanied by explanatory text. This text was informed by the literature with the highest levels of evidence, input from preceding panel deliberations (in other words, round 2 from round 1 and round 3 from rounds 1 and 2), and the teams' collective clinical and research experiences (3). The current study was reviewed by the institutional review board of the Children's Hospital of Philadelphia and exemption was granted on December 30, 2024 (IRB 24-022920).

Abbreviations

LM = lateral meniscus, MM = medial meniscus

Summary

Through a Delphi process, all 33 panelists reached consensus on topics relating to MRI reporting of knee meniscus lesions, demonstrating that interdisciplinary and multi-institutional standardization is possible.

Key Results

- Between rounds 2 and 3, agreement increased among both orthopedic surgeons and radiologists on the topics of possible tears (to express uncertainty) (73% [median score, 7] vs 94% [median score, 8]; 97% [median score, 8] vs 97% [median score, 9]; P values, .01 and .04, respectively) and tear descriptors (for treatment planning) (85% [median score, 8] vs 100% [median score, 9]; $P = .003$).
- Agreement increased among the orthopedic surgeons on the MRI criteria for tears ($P = .03$) and assessment of the postoperative meniscus (P value range, .02–.04), specifically on the reporting of equivocal findings.
- Agreement increased among the radiologists on the topics of tear patterns ($P = .04$), intrameniscal descriptors ($P = .003$), meniscus roots ($P = .03$), and re-tear descriptors ($P = .04$).

Delphi Rounds

REDCap (version 15.9.3; Research Electronic Data Capture) and Qualtrics (2024) (Qualtrics XM Platform) were used for the project. The conclusion was predetermined to occur after a maximum of three Delphi rounds. In round 1, each panelist responded to open-ended questions with free-text responses. In rounds 2 and

3, all new and updated statements, respectively, were voted on by all panel members using a 9-point Likert scale. Consensus was defined as a score of 7 or higher from at least 80% of the panelists, along with a median score of 7 or higher and an IQR of 3 or less. A 4th-year medical student (V.S.P.) with over 3 years of research experience in musculoskeletal imaging and orthopedic research managed all collected responses, ensuring anonymity by removing personal identifiers from the submissions.

Statistical Analysis

Statistical analysis was performed using R and R Studio (version 1.1.423; R Foundation for Statistical Computing). $P < .05$ was considered statistically significant. The Shapiro-Wilk test was used to determine the normality of the distribution. Continuous variables, if normally distributed, are presented as means and SDs; if not normally distributed, they are presented as medians and IQRs. Categorical variables are presented as counts and percentages. Each statistical test was selected based on the type and distribution of the variables and the number of groups. χ^2 and Mann-Whitney U analyses were used to compare agreement scores between rounds and between orthopedic surgeons and radiologists.

Results

All 33 panelists completed all three rounds of the Delphi process. A total of 32 statements were proposed by the panelists, and all achieved consensus.

Table 1: Distribution of Professional Backgrounds Reported by Delphi Panelists

Characteristic	Orthopedic Surgeons ($n = 11$)	Radiologists ($n = 22$)	P Value
Years in practice*	14 (10–23) [0–33]	23 (10–28) [6–35]	.26
Geographic location			.13
Northeast	3/11 (27.3)	11/22 (50.0)	
Midwest	5/11 (45.4)	2/22 (9.1)	
South	0/11 (0)	2/22 (9.1)	
West	3 (27.3)	6/22 (27.3)	
Canada	0/11 (0)	1/22 (4.5)	
Clinical practice			>.99
Teaching hospital	10/11 (90.9)	18/22 (81.8)	
Private practice	0/11 (0)	1/22 (4.5)	
Hybrid	1/11 (9.1)	3/22 (13.7)	
Patient population			.02†
All ages	6/11 (54.5)	6/22 (27.3)	
Predominantly pediatric	3/11 (27.3)	1/22 (4.5)	
Predominantly adult	2/11 (18.2)	15/22 (68.2)	
Meniscus cases per week			.001†
1–5	1/11 (9.1)	0/22 (0)	
6–10	6/11 (54.5)	1/22 (4.5)	
11–15	2/11 (18.2)	6/22 (27.3)	
>15	2/11 (18.2)	15/22 (68.2)	

Note.—Unless otherwise noted, values are numbers of panelists, with percentages in parentheses. Mann-Whitney U analysis was used to compare years in practice. χ^2 analysis was used to compare geographic location, clinical practice, patient population, and meniscus cases per week.

* Numbers in parentheses are IQRs, and numbers in brackets are ranges.

† Indicates statistical significance.

Panelists

Table 1 summarizes the professional backgrounds reported by the panelists. Differences were observed in both patient population and case volume between orthopedic surgeons and radiologists. Orthopedic surgeons were more likely to treat patients across all age groups (54.5% [six of 11]), whereas radiologists predominantly evaluated adult patients (68.2% [15 of 22]; $P = .02$). Moreover, orthopedic surgeons more commonly encountered 6–10 meniscus cases per week (54.5% [six of 11]), whereas radiologists more frequently interpreted more than 15 meniscus cases weekly (68.2% [15 of 22]; $P = .001$).

Delphi Rounds

In round 1, a total of 350 free-text entries (mean, 31.8 entries per topic $\pm$ 2.2 [SD]) were submitted. In round 2, 32 statements (mean, 2.9 entries per topic $\pm$ 1.6) were scored, and 75% (24 of 32) achieved consensus. By round 3, all updated statements achieved consensus. Between rounds 2 and 3, agreement increased significantly among both orthopedic surgeons and radiologists regarding statements on possible tears (statement 2.1: 73% [24 of 33 panelists; median score, 7; IQR, 6–8] vs 94% [31 of 33 panelists; median score, 8; IQR, 7–9], $P = .01$; statement 2.2: 97% [32 of 33 panelists; median score, 8; IQR, 7–9] vs 97% [32 of 33 panelists; median score, 9; IQR, 8–9], $P = .04$) and tear descriptors (statement 4.1: 85% [28 of 33 panelists; median score, 8; IQR, 7–9] vs 100% [33 of 33 panelists; median score, 9; IQR, 9–9], $P = .003$) (Tables 2–4).

Between rounds 2 and 3, agreement increased among orthopedic surgeons on the topics of MRI criteria for tears (statement 1.1: 100% [11 of 11 orthopedic surgeons; median score, 8; IQR, 8–9] vs 100% [11 of 11 orthopedic surgeons; median score, 9; IQR, 9–9], $P = .03$) and the assessment of postoperative meniscus (statement 9.3: 82% [nine of 11 orthopedic surgeons; median score, 7; IQR, 7–8] vs 100% [11 of 11 orthopedic surgeons; median score, 8; IQR, 8–9], $P = .02$; statement 9.4: 55% [six of 11 orthopedic surgeons; median score, 7; IQR, 5–8] vs 91% [10 of 11 orthopedic surgeons; median score, 8; IQR, 7–9], $P = .04$) (Table S1). Similarly, agreement increased among the radiologists on the topics of tear patterns (statement 3.1: 68% [15 of 22 radiologists; median score, 9; IQR, 6–9] vs 100% [22 of 22 radiologists; median score, 9; IQR, 9–9], $P = .04$), intrameniscal descriptors (statement 5.1: 68% [15 of 22 radiologists; median score, 8; IQR, 6–9] vs 100% [22 of 22 radiologists; median score, 9; IQR, 8–9], $P = .003$), meniscal roots (statement 6.4: 73% [16 of 22 radiologists; median score, 8; IQR, 6–9] vs 95% [21 of 22 radiologists; median score, 9; IQR, 8–9], $P = .03$), and retear descriptors (statement 10.1: 77% [17 of 22 radiologists; median score, 8; IQR, 7–9] vs 95% [21 of 22 radiologists; median score, 9; IQR, 8–9], $P = .04$) (Table S2).

Discussion

Although there are shared definitions and terminologies for describing meniscus lesions on knee MRI, controversies and differences exist between orthopedic surgeons and musculoskeletal radiologists. In three rounds of a Delphi consensus process, panel agreement improved between rounds 2 and 3, resulting in interdisciplinary consensus for all 32 consensus-based statements (Tables 2–4).

Topic 1: Criteria for MRI Diagnosis of Meniscus Tears (Unoperated Meniscus)

Statement 1.1: The primary MRI criteria for diagnosing a meniscus tear include (a) an abnormal increase in signal intensity that extends to the surface of the meniscus and (b) morphologic deformity. Specifically, diagnostic accuracy is high when linear high signal extends unequivocally to the meniscus surface on two or more slices (eg, two consecutive slices in the same imaging plane or slices in two different imaging planes) (4).

Statement 1.2: Morphologic deformity in the meniscus (eg, truncation) in an unoperated meniscus is often associated with displaced meniscus tears. Free edge blunting may result from a radial tear or from degeneration with fraying.

Since the initial use of MRI for diagnosing meniscus tears 40 years ago, the overall diagnostic performance has not substantially changed. A recent meta-analysis of 75 publications found pooled overall sensitivity and specificity of 91.0% and 87.7% for medial meniscus (MM) tears and 78.5% and 94.0% for lateral meniscus (LM) tears, respectively (5). Although this study found no significant differences in sensitivity or specificity between studies using one or more versus two or more slices for diagnosing meniscus tears, considering both the literature and clinical experience, the recommendation is to continue using the “two-slice-touch” rule for diagnosing more definitive meniscus tears in patients who have not undergone previous surgery.

Topic 2: Criteria for MRI Diagnosis of Possible Tears (Unoperated Meniscus)

Statement 2.1: Terms expressing uncertainty, such as “equivocal,” may be reported in the following circumstances: (a) abnormal morphology or intrameniscal signal contacting the meniscal articular surface on only one slice, (b) intrameniscal signal that is neither clearly confined to the substance nor definitively extending to the meniscal surface, (c) chondrocalcinosis, and (d) artifact or technical limitations (4,6–10).

Statement 2.2: Indeterminate findings may not indicate a surgically treatable lesion; however, using terms of uncertainty can provide flexibility in management for clinical providers.

When MRI findings of a meniscus tear are indeterminate, descriptors such as “possible,” “probable,” and “equivocal” are used to indicate uncertainty. In instances where meniscal abnormality is present on only one slice, the likelihood of a tear at arthroscopy is 40%–67% for the MM and 0%–33% for the LM (4,6–8). These terms of uncertainty are intended to allow flexibility in patient management and indicate that some imaging findings are equivocal for the diagnosis of a tear. In equivocal cases, the presence of associated findings (eg, parameniscal cysts, bone or joint line edema-like signal, meniscus extrusion) can increase diagnostic confidence (11).

Topic 3: Meniscus Tear Patterns

Statement 3.1: Reportable meniscus tear patterns include horizontal, longitudinal-vertical, radial, complex, bucket-handle, vertical flap, and horizontal flap tears (Fig 1).

Statement 3.2: No absolute prohibited terms for describing meniscus tear patterns have been identified.

Two prior efforts to standardize terminology include the Multicenter Orthopaedic Outcomes Network (MOON) and

Table 2. Consensus Statements on Meniscal Tear Evaluation with Agreement Scores

Topic and Statement No.	Statement	Agreement Score (%)*		
		Round 2	Round 3	P Value
MRI criteria for tears				
1.1	The primary MRI criteria for diagnosing a meniscus tear include (a) an abnormal increase in signal intensity that extends to the surface of the meniscus and (b) morphologic deformity. Specifically, diagnostic accuracy is high when linear high signal extends unequivocally to the meniscus surface on two or more slices (eg, two consecutive slices in the same imaging plane or slices in two different imaging planes).	94 (31/33) [9; 8–9]	97 (32/33) [9; 9–9]	.05
1.2	Morphologic deformity in the meniscus (eg, truncation) in an unoperated meniscus is often associated with displaced meniscus tears. Free edge blunting may result from a radial tear or from degeneration with fraying.	91 (30/33) [9; 7–9]	100 (33/33) [9; 9–9]	.11
Possible tears				
2.1 [†]	Terms expressing uncertainty, such as “equivocal,” may be reported in the following circumstances: (a) abnormal morphology or intrameniscal signal contacting the meniscal articular surface on only one slice; (b) intrameniscal signal that is neither clearly confined to the substance nor definitively extending to the meniscal surface; (c) chondrocalcinosis; and (d) artifact or technical limitations.	73 (24/33) [7; 6–8]	94 (31/33) [8; 7–9]	.01 [‡]
2.2	Indeterminate findings may not indicate a surgically treatable lesion; however, using terms of uncertainty can provide flexibility in management for clinical providers.	97 (32/33) [8; 7–9]	97 (32/33) [9; 8–9]	.04 [‡]
Tear patterns				
3.1 [†]	Reportable meniscus tear patterns include horizontal, longitudinal-vertical, radial, complex, bucket-handle, vertical flap, and horizontal flap tears.	76 (25/33) [9; 7–9]	100 (33/33) [9; 8–9]	.03 [‡]
3.2 [†]	No absolute prohibited terms for describing meniscus tear patterns have been identified.	64 (21/33) [7; 5–9]	82 (27/33) [8; 7–9]	.34
Tear descriptors				
4.1	The location of a meniscus tear should be described using the terms <i>anterior or posterior horn</i> , <i>anterior or posterior root</i> , and <i>body</i> , along with an indication of the tear’s location along the radial axis of the meniscus. This can be achieved by using the terms <i>peripheral</i> , <i>middle third</i> , or <i>free edge</i> , or by categorizing the zones into outer, middle, and inner thirds. Additionally, describing a lateral meniscus tear in relation to the popliteus hiatus can be useful.	85 (28/33) [8; 7–9]	100 (33/33) [9; 9–9]	.003 [‡]
4.2	Longitudinal-vertical and radial tears should be described as partial or complete. Optionally, an estimate of the tear size in millimeters can be reported, specifying length for longitudinal-vertical tears and width for radial tears.	85 (28/33) [8; 7–9]	94 (31/33) [9; 8–9]	.14
4.3	Key image numbers may be included in MRI reports to communicate selected findings, such as subtle tears, displaced flaps, or free edge truncation.	100 (33/33) [9; 9–9]	100 (33/33) [9; 8–9]	.81
4.4	The terms <i>stable</i> , <i>unstable</i> , <i>reparable</i> , and <i>unreparable</i> should be avoided in MRI reports.	100 (33/33) [9; 9–9]	...	...
Intrameniscal descriptors				
5.1 [†]	A descriptive delineation of intrameniscal signal (eg, globular, linear) can indicate histopathological disruption of the meniscal structure. However, these observations do not necessarily indicate a tear and cannot reliably predict the occurrence of a future tear. Additionally, intrameniscal signal in the periphery of the meniscus may reflect normal vascularity, particularly in pediatric knees.	76 (25/33) [8; 7–9]	100 (33/33) [9; 8–9]	.001 [‡]
5.2	A single time-point MRI examination does not reliably determine the chronicity of a tear.	94 (31/33) [8; 7–9]	...	...

* Numbers in parentheses are numbers of panelists. Numbers in brackets are medians and IQRs. Mann-Whitney *U* analysis was used to compare agreement scores.

[†] Statements that did not reach consensus in round 2.

[‡] Indicates statistical significance.

* Numbers in parentheses are numbers of panelists. Numbers in brackets are medians and IQRs. Mann-Whitney *U* analysis was used to compare agreement scores.

[†] Statements that did not reach consensus in round 2.

[‡] Indicates statistical significance.

Table 3. Consensus Statements on Meniscus Root, Meniscocapsular Junction, and Extrusion with Agreement Scores

Topic and Statement No.	Statement	Agreement Score (%)*		
		Round 2	Round 3	P Value
Meniscal roots				
6.1 [†]	The meniscal roots anchor the anterior and posterior horns of the meniscus to the intercondylar region of the tibial plateau. The meniscal root functionally includes the enthesal attachment and approximately 9–10 mm of the adjacent meniscus.	79 (26/33) [8; 7–9]	100 (33/33) [9; 8–9]	.09
6.2	Root injuries include avulsions and radial tears. Avulsions occur when there is a detachment of the root's enthesal attachment, often with a small bone fragment. Radial tears may be partial, involving the free edge, or complete, extending to the periphery.	85 (28/33) [8; 8–9]	88 (29/33) [8; 8–9]	.64
6.3 [†]	When meniscus tears involving the horns extend into the root, describing the tear morphology and extent of the tear may be helpful.	79 (26/33) [8; 7–9]	97 (32/33) [8; 8–9]	.04 [‡]
6.4	Intrasubstance signal of the root may be observed but should not be defined as a root tear.	82 (27/33) [8; 7–9]	97 (32/33) [9; 8–9]	.04 [‡]
Posteromedial meniscocapsular junction				
7.1	Meniscus ramp lesions refer to injuries associated with the posteromedial meniscocapsular junction. These include tears involving the meniscocapsular attachment(s), the meniscotibial attachment(s), and the periphery of the MM posterior horn.	88 (29/33) [9; 7–9]	...	...
7.2	MRI findings associated with arthroscopically confirmed meniscus ramp lesions include vertically oriented fluid-like signal at the meniscocapsular junction or in the peripheral third of the meniscus (longitudinal-vertical tears), as well as contour irregularity at the periphery of the MM posterior horn. Additionally, bone marrow edema-like signal at the posteromedial tibial plateau is a common but nonspecific secondary finding.	97 (32/33) [8; 7–9]	...	...
Meniscus extrusion				
8.1	MRI of the knee is the preferred imaging modality for evaluating suspected meniscus extrusion. Axial loading during imaging has the potential to enhance the evaluation of meniscus dysfunction, but it has not yet been translated into routine clinical practice.	97 (32/33) [8; 8–9]	...	...
8.2	The standard evaluation for meniscus extrusion on MRI is performed on a coronal slice that captures the greatest volume of the medial tibial spine.	97 (32/33) [9; 8–9]	...	...
8.3	Reference points for measuring meniscus extrusion include the peripheral chondroosseous margin or the transition point of the tibial plateau from horizontal to vertical, excluding osteophytes, and the outer edge of the meniscus. Current cutoff values for pathological extrusion are ≥ 3 mm for the MM and ≥2 mm for the LM.	82 (27/33) [8; 7–9]	...	...
8.4	Sonography can detect and reproducibly measure MM extrusion; however, data on LM extrusion are limited.	94 (31/33) [8; 7–9]	97 (32/33) [9; 8–9]	.14

Note.—LM = lateral meniscus. MM = medial meniscus.

* Numbers in parentheses are numbers of panelists. Numbers in brackets are medians and IQRs. Mann-Whitney *U* analysis was used to compare agreement scores.

[†] Statements that did not reach consensus in round 2.

[‡] Indicates statistical significance.

the International Society of Arthroscopy, Knee Surgery and Orthopaedic Sports Medicine (ISAKOS). Both have demonstrated substantial interobserver agreement (12,13). Although similarities exist between these classifications, they are not identical, and neither has gained universal adoption. In some practices,

additional terms continue to be used to describe tears, such as *oblique*, *parrot-beak*, *macerated*, and *multidirectional*. The term *oblique* has been applied to two distinctly different tear patterns (vertical flap and obliquely oriented horizontal tears) and should therefore be used with caution.

Table 4. Consensus Statements on Postoperative Meniscus and Ancillary Findings with Agreement Scores

Topic and Statement No.	Statement	Agreement Score (%)*		
		Round 2	Round 3	P Value
Assessment of postoperative meniscus				
9.1	MRI without contrast material is the preferred imaging modality for assessing suspected meniscus retears following partial meniscectomy or meniscus repair.	88 (29/33) [8; 7–9]	...	...
9.2 [†]	MRI findings indicative of reter include (a) abnormal meniscal morphology not explained by prior surgery, (b) a displaced meniscus flap or fragment, (c) linear fluid-like signal intensity extending to the meniscal surface, (d) intermediate signal extending to the surface on images showing an increased gap at the site of the tear compared to the preoperative MRI, (e) diastasis at the repair site greater than 1 mm, and (f) the presence of a radial tear.	79 (26/33) [8; 7–8]	91 (30/33) [8; 8–8]	.04 [‡]
9.3	Equivocal findings of reter on MRI include linear intermediate signal extending to the surface on any pulse sequence in the absence of additional data (eg, comparison MRI or operative note).	85 (28/33) [8; 7–8]	91 (30/33) [8; 7–9]	.23
9.4 [†]	MR arthrography may be considered if routine MRI findings are equivocal, if a prior MRI is unavailable for comparison, or if prior operative reports are not accessible. The selection of imaging may depend on the orthopedic surgeon's preference and/or local practice patterns, as the decision to reoperate may be based on clinical findings.	70 (23/33) [7; 6–8]	91 (30/33) [8; 7–9]	.01 [‡]
9.5	Findings of reter on MR arthrography, include: (a) contrast material extending into the meniscus, (b) linear high signal on T2-weighted images extending to the surface, and (c) a displaced meniscus flap or fragment.	82 (27/33) [8; 7–8]	...	...
9.6	CT arthrography may also be considered in the evaluation of the postoperative meniscus.	88 (29/33) [8; 7–9]	97 (32/33) [9; 8–9]	.41
Reter descriptors				
10.1	Morphologic and signal intensity descriptors are recommended for MRI reporting of the postoperative meniscus. For morphology, any descriptive terms for meniscus tears may be used according to their definitions, such as “displaced flap” and its anatomic location. For signal intensity, descriptions should specify the intensity (eg, high [fluid-like], intermediate [articular cartilage-like]) and whether there is surface extension. Postoperative findings can include details about suture tracts, fixation devices, and types of tibial fixation.	85 (28/33) [8; 7–9]	97 (32/33) [8; 8–9]	.04 [‡]
10.2	The term <i>residual tear</i> and the numerical grading of intrameniscal signal changes are discouraged for MRI reporting purposes.	91 (30/33) [9; 8–9]	...	...
Ancillary findings				
11	Findings associated with meniscus tears should generally be reported due to their potential treatment implications and their relevance for preoperative planning. These findings include (a) meniscus extrusion, (b) subchondral marrow edema, (c) disruption of perimeniscal attachments, (d) discoid meniscus and associated findings, and (e) parameniscal cyst(s).	91 (30/33) [9; 8–9]	97 (32/33) [9; 8–9]	.56

^{*} Numbers in parentheses are numbers of panelists. Numbers in brackets are medians and IQRs. Mann-Whitney *U* analysis was used to compare agreement scores.

[†] Statements that did not reach consensus in round 2.

[‡] Indicates statistical significance.

Topic 4: Meniscus Tear Descriptors

Statement 4.1: The location of a meniscus tear should be described using the terms *anterior or posterior horn*, *anterior or posterior root*, and *body*, along with an indication of the tear's location along the radial axis of the meniscus. This can be achieved by using the terms *peripheral*, *middle third*, or *free edge*, or by categorizing the zones

into outer, middle, and inner thirds. Additionally, describing a lateral meniscus tear in relation to the popliteus hiatus can be useful.

Statement 4.2: Longitudinal-vertical and radial tears should be described as partial or complete. Optionally, an estimate of the tear size in millimeters can be reported, specifying length for longitudinal-vertical tears and width for radial tears.

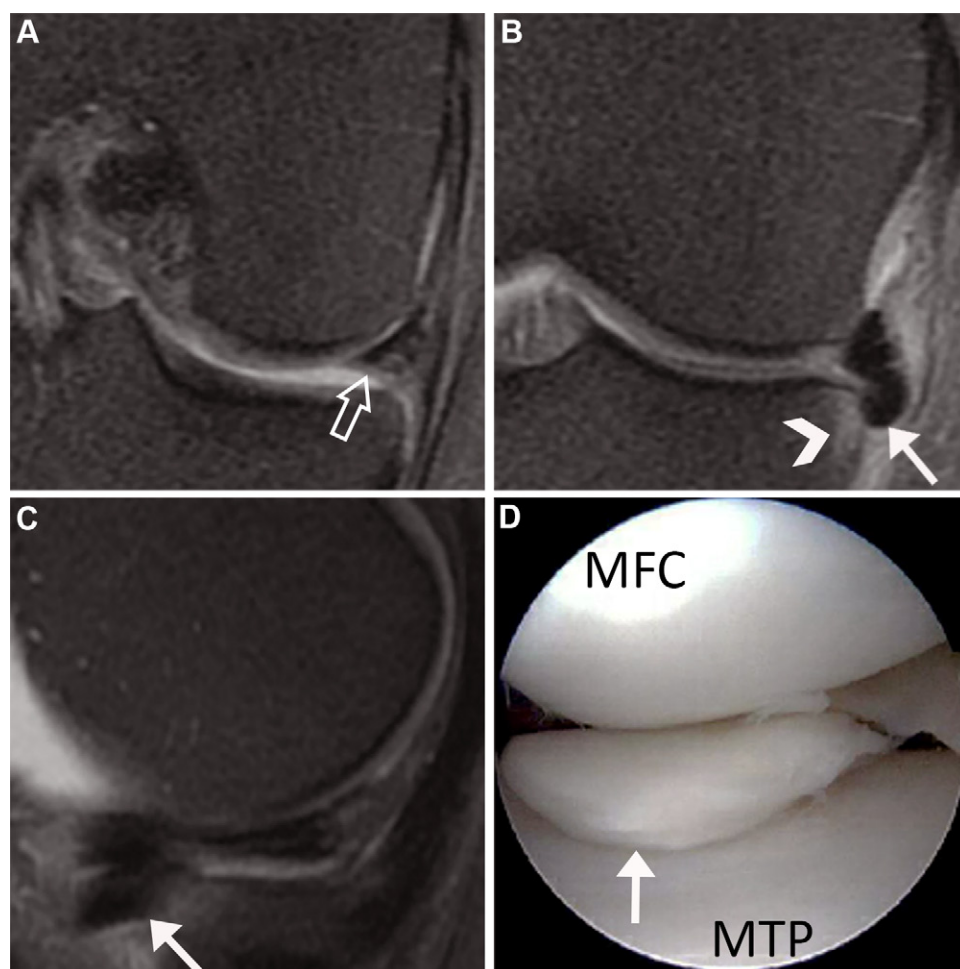


Figure 1: Horizontal flap tear in a 39-year-old woman with knee pain and locking. (**A, B**) Coronal proton density-weighted fat-suppressed and (**C**) sagittal T2-weighted fat-suppressed MRI scans reveal a horizontal tear (arrow in **A**) with decreased volume inferiorly, which serves as the donor site for the displaced flap located in the medial meniscotibial gutter (arrow in **B** and **C**). Note the associated edema-like signal in the underlying subchondral bone marrow (chevron in **B**) and in the adjacent parameniscal soft tissue. (**D**) Arthroscopic view shows the meniscus flap (arrow) retrieved into the joint. MFC = medial femoral condyle, MTP = medial tibial plateau.

Statement 4.3: Key image numbers may be included in MRI reports to communicate selected findings, such as subtle tears, displaced flaps, or free edge truncation.

Statement 4.4: The terms *stable*, *unstable*, *reparable*, and *unreparable* should be avoided in MRI reports.

The location of a meniscus tear is best described using unambiguous anatomic terms, which are generally preferred over the use of red (vascularized), red-white, or white (non-vascularized) zones. This is because meniscus vascularity may vary among individuals, including with age, and is more accurately accessed through surgery than MRI (13). In some cases, it may be beneficial to further specify the location of peripheral longitudinal-vertical tears by measuring the distance to the meniscocapsular junction, using the term *meniscus ramp lesion* for a tear at the MM periphery or meniscocapsular junction, or describing the location of an LM tear in relation to the popliteus hiatus, as these factors influence management decisions (12–15). Radiologists and orthopedic surgeons commonly report tears as extending to the superior (femoral) surface, inferior (tibial) surface, and/or free edge of the meniscus. Longitudinal-vertical tears should be referred to as complete (full-thickness) when they extend to both

the superior and inferior surfaces of the meniscus, and as partial or incomplete if they only extend to one surface. For horizontal and radial tears, tears are described as complete (full-width) when they extend from (or near) the free edge to the periphery of the meniscus, while partial or incomplete tears involve less than the entire width of the meniscus. The completeness of a tear is one factor that affects its stability and potential for repair (Fig 2).

A meniscus can be described as having decreased volume if it is unclear whether the appearance is due to a tear or a prior partial meniscectomy. Additionally, it may be useful to include image numbers when describing subtle meniscus findings or the location of a displaced flap or fragment. Optionally, an indication of its size, either on a relative scale (small, medium, or large) or by measuring its dimensions, can also be included.

Topic 5: Intrameniscal Signal Descriptors

Statement 5.1: A descriptive delineation of intrameniscal signal (eg, globular, linear) can indicate histopathologic disruption of the meniscal structure (16). However, these observations do not necessarily indicate a tear and cannot

reliably predict the occurrence of a future tear. Additionally, intrameniscal signal in the periphery of the meniscus may reflect normal vascularity, particularly in pediatric knees.

Statement 5.2: A single-time-point MRI examination does not reliably determine the chronicity of a tear.

On MRI, signal alteration confined within the meniscus (not extending to the surface) can be incidental and arise from various etiologies, including peripheral vascularity, contusion, degeneration, chondrocalcinosis, ossification (eg, meniscal ossicle), and intrameniscal cysts (11,16–18). In cases of meniscus contusion, intrameniscal signal in the periphery (the vascularized part) of the meniscus is often described as “amorphous” and regresses over time (17). While punctate and globular intrameniscal signal rarely progresses to a tear, linear intrameniscal signal is associated with an increased risk of a tear within the same meniscal segment (19).

In general, terms that suggest etiology (traumatic vs degenerative) or duration (acute vs chronic) should be avoided, as the criteria for making these determinations based on MRI findings are unreliable, particularly in the absence of prior imaging or clinical context.

Topic 6: Meniscal Root Definition and Tears

Statement 6.1: The meniscal roots anchor the anterior and posterior horns of the meniscus to the intercondylar region of the tibial plateau. The meniscal root functionally includes the enthesal attachment and approximately 9–10 mm of the adjacent meniscus (20–22).

Statement 6.2: Root injuries include avulsions and radial tears. Avulsions occur when there is a detachment of the root's enthesal attachment, often with a small bone fragment. Radial tears may be partial, involving the free edge, or complete, extending to the periphery (23–25).

Statement 6.3: When meniscus tears involving the horns extend into the root, describing the tear morphology and extent of the tear may be helpful.

Statement 6.4: Intrameniscal signal of the root may be observed but should not be defined as a root tear (25,26).

Root tears increase the risk of early and progressive osteoarthritis (27). These tears are more likely to affect the posterior roots than the anterior roots (Fig 3). Posterior root tears involving the LM are most common in younger males following trauma, while those involving the MM are more frequent in middle-aged to older females with a high body mass index (26,28–30). Root tears include not only tears at the enthesal attachments but also as tears involving the junctional tissue contiguous with the horns (23,24,26,29,31). Although various classification systems exist, no single system has achieved universal adoption (23,24). Therefore, MRI reporting should be descriptive and include location, morphology, and extent of root tears as well as any

associated findings (eg, cartilage lesions, bone marrow edema-like signal, osseous avulsions, meniscus extrusion, and meniscofemoral ligament tear).

Topic 7: Posteromedial Meniscocapsular Junction Definition and Tears

Statement 7.1: Meniscus ramp lesions refer to injuries associated with the posteromedial meniscocapsular junction. These include tears involving the meniscocapsular attachment(s), the meniscotibial attachment(s), and the periphery of the MM posterior horn (14,15,32).

Statement 7.2: MRI findings associated with arthroscopically confirmed meniscus ramp lesions include vertically oriented

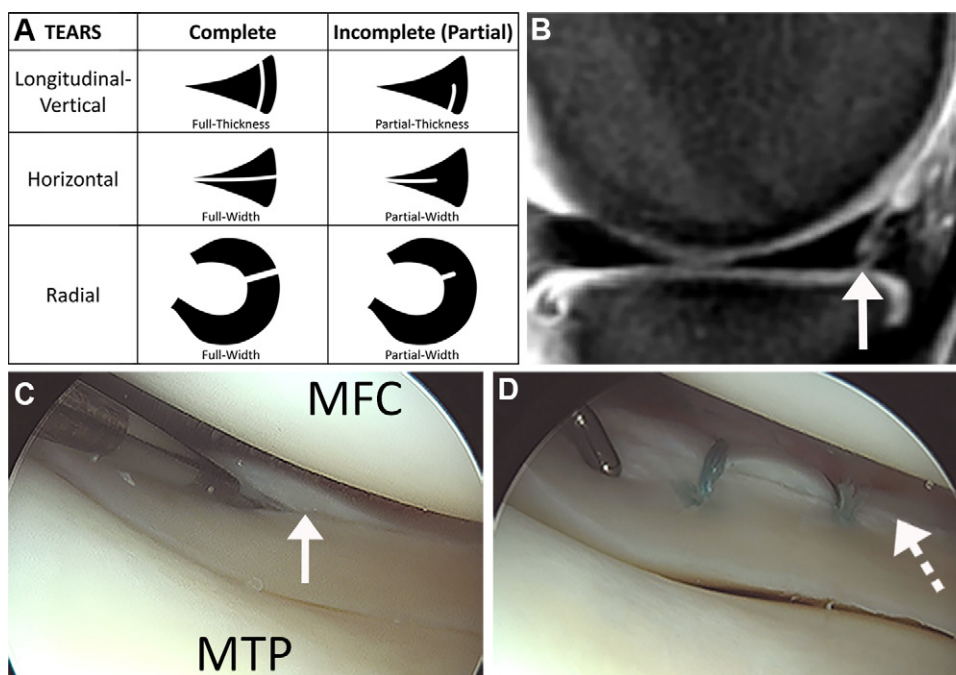


Figure 2: Complete versus incomplete (partial) meniscus tears. (A) Illustration shows that complete longitudinal-vertical tears extend to both the superior and inferior surfaces of the meniscus, while complete horizontal and radial tears at (or near) the free edge that extend to the periphery of the meniscus. In contrast, incomplete longitudinal-vertical tears extend to only one surface, and incomplete horizontal and radial tears involve less than the entire width of the meniscus. In a 15-year-old boy with an anterior cruciate ligament tear, (B) sagittal T2-weighted fat-suppressed MRI scan reveals a complete (full-thickness) longitudinal-vertical tear in the posterior horn (arrow). (C, D) Arthroscopic views with probing document the tear (arrow in C) and the subsequent suture repair (arrow in D). MFC = medial femoral condyle, MTP = medial tibial plateau.

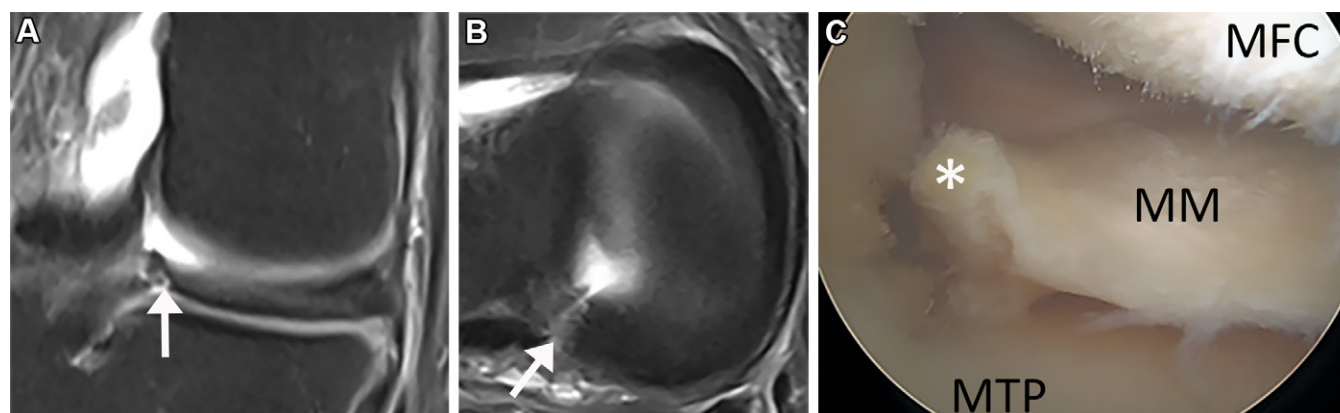


Figure 3: Medial meniscus complete posterior root tear in a 69-year-old man with knee pain. (A) Coronal proton density-weighted fat-suppressed and (B) axial T2-weighted fat-suppressed MRI scans reveal a complete radial tear at the posterior root (arrow). (C) Arthroscopic view confirms the torn root stump (*) and shows chondrosis with fraying over the medial femoral condyle (MFC). MM = medial meniscus, MTP = medial tibial plateau.

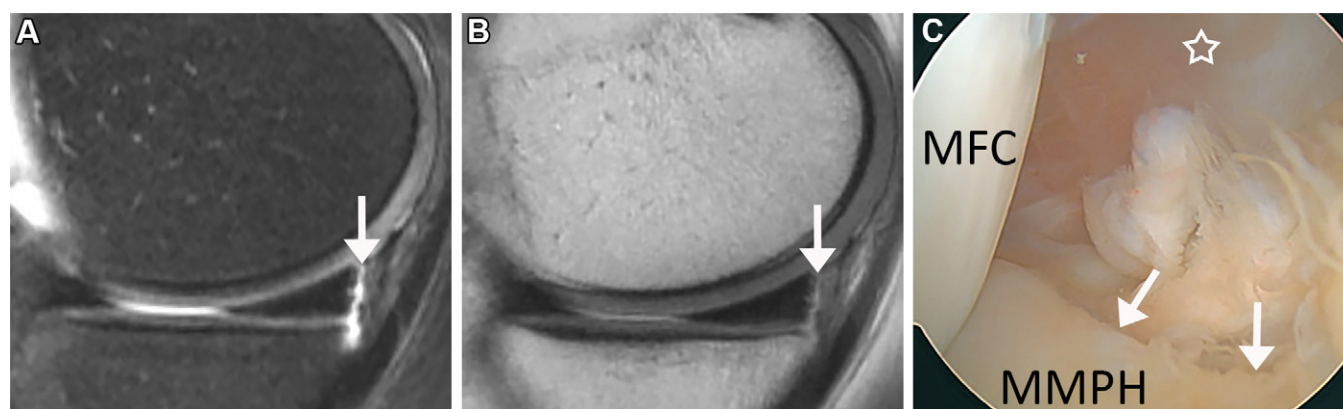


Figure 4: Posteromedial meniscocapsular tear (ramp lesion) in a 21-year-old woman with a history of anterior cruciate ligament reconstruction and a recent soccer injury. Sagittal **(A)** T2-weighted fat-suppressed and **(B)** proton density-weighted MRI scans reveal full-thickness fluid-like signal intensity at the meniscocapsular junction (arrow). **(C)** Arthroscopic transcondylar view confirms the separation (arrows) between the medial meniscus posterior horn (MMPH) and the posterior joint capsule (star). MFC = medial femoral condyle.

fluid-like signal at the meniscocapsular junction or in the peripheral third of the meniscus (longitudinal-vertical tears), as well as contour irregularity at the periphery of the MM posterior horn (33–36). Additionally, bone marrow edema-like signal at the posteromedial tibial plateau is a common but nonspecific secondary finding (37).

Risk factors for meniscus ramp lesions include a complete anterior cruciate ligament tear, age under 30 years, and male sex, with pooled odds ratios of 3.0, 2.02, and 1.58, respectively (37). When using arthroscopy as the reference standard for diagnosis, MRI in pediatric and adult patients has an overall sensitivity of 50%–71% and specificity of 75%–94%. The lower diagnostic performance in pediatric patients is attributed to increased peripheral vascularization and a more robust healing potential (38–40). A qualitative description of injuries involving the posteromedial meniscocapsular junction is recommended over strict adherence to a specific classification, as no classification has yet garnered widespread clinical or radiologic adoption (Fig 4).

Topic 8: Meniscus Extrusion Assessment and Criteria

Statement 8.1: MRI of the knee is the preferred imaging modality for evaluating suspected meniscus extrusion. Axial loading during imaging has the potential to enhance the evaluation of meniscus dysfunction, but it has not yet been translated into routine clinical practice.

Statement 8.2: The standard evaluation for meniscus extrusion at MRI is performed on a coronal slice that captures the greatest volume of the medial tibial spine.

Statement 8.3: Reference points for measuring meniscus extrusion include the peripheral chondroosseous margin or the

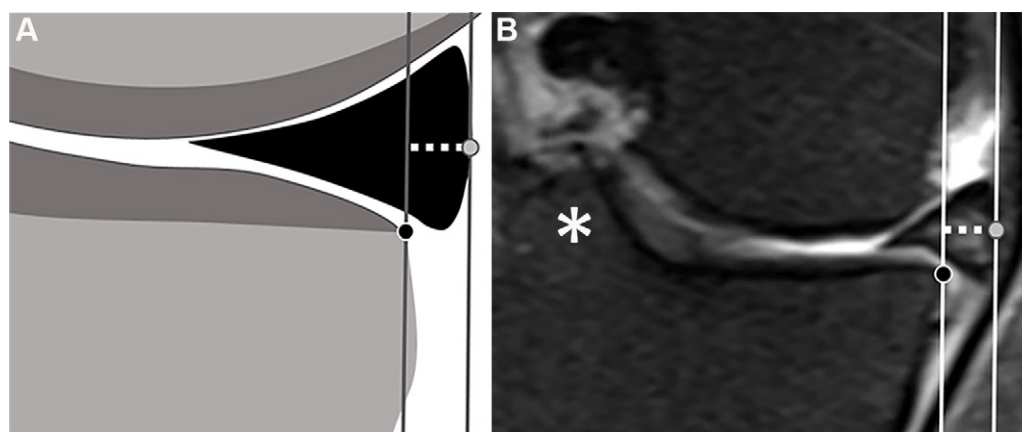


Figure 5: Landmarks for quantitative determination of extrusion of the body segment. **(A)** Illustration depicts reference points for measuring meniscus extrusion (dashed line) between the peripheral chondroosseous margin (the point where the tibial plateau transitions from horizontal to vertical, excluding osteophytes, marked by a black dot) and the outer edge of the meniscus (gray dot). **(B)** Coronal proton density-weighted fat-suppressed MRI scan in a 44-year-old woman with a complete tear of the medial meniscus posterior root (not visible in this image) shows extrusion of the body segment (dashed line) at the section depicting greatest volume of the medial tibial spine (*).

transition point of the tibial plateau from horizontal to vertical, excluding osteophytes, and the outer edge of the meniscus. Current cutoff values for pathologic extrusion are 3 mm or more for the MM and 2 mm or more for the LM (41–44).

Statement 8.4: Sonography can help detect and reproducibly measure MM extrusion (45); however, data on LM extrusion are limited.

In a retrospective cross-sectional study that included more than 1500 subjects, meniscus extrusion was associated with meniscus tears, cartilage damage, and knee malalignment, but not with the presence of a knee effusion (46). Specifically, tears that disrupt the hoop stress function of the meniscus, such as root, radial, and complex tears, can lead to radial expansion or meniscus extrusion (47) (Fig 5). Although loaded knee conditions and dynamic change assessment can aid diagnosis and prediction, variations in imaging protocols and the lack of reference values hinder clinical translation (48).

For the MM, a commonly used threshold is 3 mm (41–44). However, some authors have suggested using thresholds of 5 mm or 4 mm, while others have proposed relying on the relative percentage of extrusion instead of an absolute measured value (46,49,50). Data for the LM are more limited, and an absolute

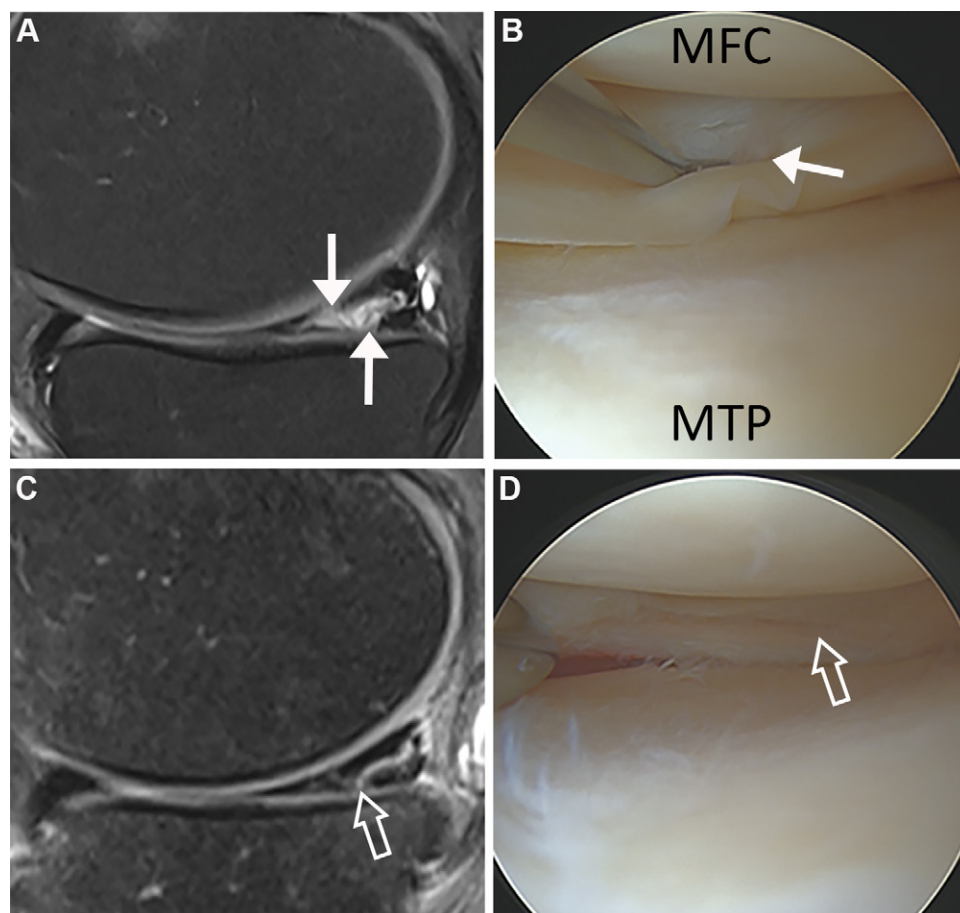


Figure 6: Retear of the posterior horn of the medial meniscus in a 39-year-old woman. **(A)** Sagittal T2-weighted fat-suppressed MRI scan and **(B)** corresponding arthroscopic view reveal a complex, predominantly horizontal tear extending to both the superior and lower surfaces (arrows), which was subsequently repaired with sutures. **(C)** Follow-up sagittal T2-weighted fat-suppressed MRI scan obtained 4 months later shows linear fluid-like signal extending to the inferior surface of the meniscus (arrow). **(D)** Repeat arthroscopic view indicates a persistent tear (arrow). MFC = medial femoral condyle, MTP = medial tibial plateau.

threshold of 2 mm has been suggested, which does not overlap with a mean body extrusion of 1.8 mm observed in middle-aged and older adults without radiographic osteoarthritis (42,43).

Topic 9: Assessment of the Postoperative Meniscus

Statement 9.1: MRI without contrast material is the preferred imaging modality for assessing suspected meniscus retears following partial meniscectomy or meniscus repair.

Statement 9.2: MRI findings indicative of re-tear include (a) abnormal meniscal morphology not explained by prior surgery, (b) a displaced meniscus flap or fragment, (c) linear fluid-like signal intensity extending to the meniscal surface, (d) intermediate signal extending to the surface on images showing an increased gap at the site of the tear compared with the preoperative MRI, (e) diastasis at the repair site greater than 1 mm, and (f) the presence of a radial tear (51–53) (Fig 6).

Statement 9.3: Equivocal findings of re-tear on MRI include linear intermediate signal extending to the surface on any pulse sequence in the absence of additional data (eg, comparison MRI or operative note) (51,54–56).

Statement 9.4: MR arthrography may be considered if routine MRI findings are equivocal, if a prior MRI is unavailable for comparison, or if prior operative reports are not accessible. The

selection of imaging may depend on the orthopedic surgeon's preference and/or local practice patterns, as the decision to reoperate may be based on clinical findings.

Statement 9.5: Findings of re-tear at MR arthrography, include (a) contrast material extending into the meniscus, (b) linear high signal on T2-weighted images extending to the surface, and (c) a displaced meniscus flap or fragment (54–57).

Statement 9.6: CT arthrography may also be considered in the evaluation of the postoperative meniscus (58).

The diagnostic performance of MRI for retears involving the postoperative meniscus is lower compared with tears involving native menisci (58). This reduced performance has been attributed to the presence of preexisting intrameniscal degeneration and/or postoperative fibrovascular granulation tissue. While smaller resections demonstrate comparable performance to untreated menisci, larger resections (>25%) and repaired menisci require modified MRI criteria to improve diagnostic performance (54–57). MR arthrography modestly outperforms MRI in patients who

have undergone a larger partial meniscectomy and meniscus repair. This improvement is generally attributed to increased intraarticular pressure, which helps drive fluid into tears, particularly in the absence of joint effusion (54–57,59).

Topic 10: Meniscus Retear Descriptors

Statement 10.1: Morphologic and signal intensity descriptors are recommended for MRI reporting of the postoperative meniscus. For morphology, any descriptive terms for meniscus tears may be used according to their definitions, such as “displaced flap” and its anatomic location. For signal intensity, descriptions should specify the intensity (eg, high [fluid-like], intermediate [articular cartilage-like]) and whether there is articular surface extension. Postoperative findings can include details about suture tracts, fixation devices, and types of tibial fixation.

Statement 10.2: The term *residual tear* and the numerical grading of intrameniscal signal changes are discouraged for MRI reporting purposes.

Comparison with prior MRI and operative reports can be immensely helpful in interpreting postoperative findings (58). Reporting a tear as “residual” can imply surgical failure or incomplete treatment; thus, the use of this descriptor in MRI reports is discouraged.

Topic 11: Reportable Ancillary Findings

Statement 11: Findings associated with meniscus tears should generally be reported due to their potential treatment implications and their relevance for preoperative planning. These findings include (a) meniscus extrusion, (b) subchondral marrow edema, (c) disruption of perimeniscal attachments, (d) discoid meniscus and associated findings, and (e) parameniscal cyst(s) (11,37,60,61).

Our study has limitations. First, our consensus-based approach synthesizes and harmonizes existing evidence; by design, it does not generate new empirical data. As these consensus statements have not yet been implemented in clinical practice, their long-term impact on diagnostic accuracy, reporting consistency, patient outcomes, and workflow efficiency remains untested. We encourage prospective, longitudinal, and ideally multicenter implementation studies to evaluate effectiveness and guide iterative refinement. Second, while we included a representative panel of subspecialty experts, there was an inclusion bias as panel members were solely from North American institutions. This limitation was justified due to differences in health care systems across geographic regions.

Conclusion

Effective and efficient communication is critical for optimizing patient care. Although our study found initial differences between orthopedic surgeons and radiologists on various topics related to knee MRI reporting of meniscus lesions, all 32 consensus-based statements achieved panel consensus. This marks a crucial step toward interdisciplinary and multi-institutional standardization.

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